

Cost-Sensitive Random Forest with Savitzky-Golay Filtering for BLE Room Localization in Nursing Homes

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Abstract. We address indoor localization on a 22-room nursing floor using BLE RSSI fingerprints from 25 beacons. The real-world dataset presents two challenges: severe class imbalance (up to $407\times$) and temporal RSSI instability caused by hardware and environmental variation. We propose a cost-sensitive spatiotemporal classification algorithm that combines Savitzky–Golay signal smoothing, sliding-window RSSI descriptors, and a class-weighted Random Forest. The algorithm is designed to stabilize noisy fingerprints, distinguish rooms with similar instantaneous signals, and learn minority-room boundaries without generating synthetic samples. Stratified 5-fold cross-validation yields a weighted F1-score of 0.835 and a macro F1-score of 0.834; adding SG filtering increases macro F1 from 0.819 to 0.834. Leave-One-Day-Out evaluation gives a mean weighted F1-score of 0.600, revealing substantial performance degradation across recording days. The method requires neither synthetic data nor additional sensing modalities and therefore provides a lightweight baseline for practical deployment.

1 Introduction

The aging population in developed countries has increased demand for caregivers, making work efficiency a critical priority. Nursing homes increasingly adopt digital recording systems on smartphones and tablets to replace paper records, reduce documentation time, and enable data-driven analysis. To complement these systems and support workload analysis, reliable tracking of caregiver location is valuable.

Bluetooth Low Energy (BLE) beacons are widely deployed for indoor localization due to low cost, low power, and ease of installation [1]. However, BLE-based localization in real nursing environments faces two critical challenges. First, Received Signal Strength Indicator (RSSI) measurements suffer from severe noise:

$$\text{RSSI}(t) = P_{\text{tx}} - 10n \log_{10}(d) - \eta(t) \quad (1)$$

where P_{tx} is the transmitted power, n is the path-loss exponent, d is the distance, and $\eta(t)$ captures multipath fading, body shadowing, and device orientation [2]. Second, location data exhibits extreme class imbalance: caregivers spend most of their shift in communal areas (nurse station: 9,361 samples; cafeteria: 5,224; kitchen: 3,102), while the rarest rooms appear only dozens of times (room 516: 23 samples)—a $407\times$ imbalance ratio that heavily biases standard classifiers. The full per-room distribution is detailed in Section 4.

Synthetic oversampling techniques such as SMOTE [3] are commonly used to mitigate imbalance. In our experiments, however, SMOTE produces the largest train–validation gap among the evaluated imbalance strategies (Fig. 3).

Existing methods typically address RSSI denoising, temporal modeling, or class imbalance as separate problems. This separation is insufficient for the present setting: smoothing alone does not prevent majority-room bias, resampling does not stabilize temporally noisy fingerprints, and single-snapshot classifiers cannot exploit short-term signal structure. We therefore propose a unified algorithm for imbalanced BLE room localization. Our main contributions are:

1. **We propose a cost-sensitive spatiotemporal localization algorithm for noisy and imbalanced BLE fingerprints.** The algorithm couples signal stabilization, temporal representation, and minority-aware classification in a single processing chain rather than treating them as independent post-processing steps.
2. **We mitigate RSSI instability and single-snapshot ambiguity by denoising before temporal aggregation.** Per-beacon SG filtering suppresses short-term fluctuations, after which sliding-window statistics preserve the local variation needed to distinguish rooms with similar instantaneous RSSI patterns. This ordering increases macro F1-score from 0.8191 to 0.8336 relative to the otherwise matching unsmoothed configuration.
3. **We address the $407\times$ class imbalance without synthesizing RSSI fingerprints.** Class-dependent costs are incorporated directly into Random Forest training through balanced subsampling. This avoids the potentially non-physical interpolations introduced by SMOTE and yields a smaller train–validation gap (0.100 versus 0.146) in our experiments.

The proposed algorithm is evaluated using both stratified cross-validation and Leave-One-Day-Out analysis. The latter is reported as a robustness diagnostic rather than an algorithmic contribution. Experimental results on a real nursing-facility dataset show that configuration V3 achieves a weighted F1-score of 0.8351 and a macro F1-score of 0.8336. The full comparison across five explored configurations is presented in Section 4 (Table 2).

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes our preprocessing pipeline and classifier. Section 4 presents experimental setup and results. Section 5 concludes and discusses future directions.

2 Related Work

2.1 BLE Indoor Localization

Indoor localization in healthcare and nursing environments has garnered significant attention, with BLE emerging as a practical option because of its low cost, low power consumption, and unobtrusive deployment. Ayub et al. [4] compared machine-learning classifiers for BLE localization, and García-Paterna et al. [5] studied room-level localization based on BLE fingerprints. Such classifier-centered approaches establish the feasibility of BLE localization, but they do not jointly model short-term RSSI variation and severe room-frequency imbalance. Consequently, a classifier can fit dominant-room fingerprints while remaining sensitive to transient RSSI fluctuations in infrequently visited rooms.

Garcia and Inoue [6] addressed erroneous labels by relabeling signal patterns using stationary beacons in nursing care facilities. Relabeling can correct a sample only when the observed pattern is sufficiently characteristic of another room. In our dataset, however, room-level mean RSSI vectors are nearly indistinguishable, so cosine-similarity relabeling cannot reliably determine the correct destination class. Moreover, relabeling does not itself suppress within-room RSSI noise or compensate for class imbalance.

2.2 RSSI Smoothing and Denoising

Due to environmental interference, raw BLE RSSI signals are notoriously volatile. Common smoothing techniques include Simple Moving Average (SMA) and Kalman filters. SMA reduces variance but can flatten peaks and blur sudden RSSI shifts, while Kalman filtering requires a state-transition model that is difficult to specify for irregular human movement. Wavelet and smoothing filters have also been studied for BLE indoor localization [7]. These methods improve signal quality, but denoising alone cannot resolve rooms with similar fingerprints and does not prevent a classifier from favoring majority classes.

To reduce short-term fluctuations without replacing each neighborhood by a constant average, the Savitzky–Golay (SG) filter provides a useful alternative. It applies local polynomial regression and can retain local signal shape more effectively than a simple moving average [8]. Halder et al. [2] evaluated SG as a baseline for ZigBee RSSI smoothing, but smoothing was not integrated with temporal room descriptors and cost-sensitive multi-class learning. Our algorithm uses SG as the first stage of a broader classifier, so the stabilized sequence can subsequently be summarized over time and learned under class-dependent costs.

2.3 Class Imbalance in RSSI Data

Data collected from real-world nursing facilities inherently exhibits extreme class imbalance. Caregivers spend disproportionately large amounts of time in communal areas (e.g., nurse station, 9,361 samples) compared to individual patient rooms (e.g., room 516, 23 samples). Conventional approaches to mitigate this often rely on synthetic data generation, most notably SMOTE [3].

Although SMOTE is effective on many structured datasets, synthetic interpolation can be problematic when RSSI classes already overlap strongly because an interpolated vector may not correspond to a physically observed propagation condition. SMOTE also changes the sample distribution without reducing the temporal noise present in the original fingerprints. In our experiments, it produces the largest train–validation gap (0.1459) among the tested strategies.

We therefore use cost-sensitive learning within the localization algorithm. During tree construction, `balanced_subsample` weighting increases the influence of minority classes. Unlike SMOTE, this approach does not invent intermediate RSSI vectors; unlike relabeling, it does not require clearly separated room centroids. It yields the smallest train–validation gap (0.0997) in our comparison.

2.4 Temporal Features for Localization

Traditional BLE fingerprinting relies on single-snapshot RSSI measurements, which can be spatially ambiguous because different locations may exhibit similar instantaneous RSSI signatures. Wang et al. [9] addressed a related problem in WiFi positioning by constructing temporal fingerprint slices. Temporal context helps distinguish observations that appear similar at one instant, but the WiFi method does not address the severe class imbalance and beacon-specific noise encountered in the present BLE nursing-home dataset.

Sequence models such as LSTMs can learn temporal dependencies, but they generally require more training data and computation than tree-based models. Existing temporal approaches may also process raw or minimally filtered sequences, allowing environmental fluctuations to propagate into the temporal representation. Thus, temporal modeling alone does not solve the combined noise and imbalance problem.

In summary, the approaches reviewed above provide useful solutions to individual parts of the problem, but they do not jointly (i) stabilize each beacon sequence, (ii) construct short-term room descriptors, and (iii) learn minority classes without synthetic fingerprints. The proposed

algorithm addresses this gap by applying SG filtering before sliding-window aggregation and then training a cost-sensitive Random Forest on the resulting spatiotemporal representation.

3 Methodology

3.1 Preprocessing Pipeline

Raw BLE RSSI measurements from each timestamp are first aggregated by taking the mean across repeated detections. Missing beacon readings are filled with -100 dBm (no detection). Data from user ID 90 and label data from user ID 97 are merged via a 4-second tolerance window to account for timestamp misalignment.

Each beacon’s RSSI time series is then denoised using a Savitzky-Golay (SG) filter [8] with window length 5 and polynomial order 2. The SG filter performs local polynomial regression, preserving signal shape while removing high-frequency fluctuations:

$$\text{RSSI}_{\text{sg}}[i] = \sum_{j=-2}^2 c_j \cdot \text{RSSI}[i + j] \quad (2)$$

where c_j are least-squares convolution coefficients. This step is applied per beacon across all timestamps before temporal feature extraction.

3.2 Feature Extraction

A sliding window of $k = 5$ consecutive timestamps is applied. For each window centered at timestamp t , four statistics—mean, standard deviation, minimum, and maximum—are computed per beacon, producing $4 \times 25 = 100$ features per sample:

$$\mathbf{x}_t = [\mu_1, \sigma_1, \min_1, \max_1, \dots, \mu_{25}, \sigma_{25}, \min_{25}, \max_{25}] \quad (3)$$

This captures spatial signatures over short temporal windows, enabling discrimination of rooms with similar instantaneous RSSI patterns.

3.3 Classifier and Imbalance Handling

We train a Random Forest classifier with 100 trees, a maximum depth of 20, and `class_weight` set to `balanced_subsample`. This cost-sensitive approach increases the weight assigned to minority classes without generating synthetic data. The configuration is evaluated using 5-fold stratified cross-validation across 22 semantic room classes.

3.4 Validation Strategy

As recommended in the challenge guidelines, we employ two validation strategies:

1. **5-fold stratified CV** (in-distribution evaluation): Data is randomly shuffled and split into five folds while preserving class proportions. This protocol uses the limited 4-day dataset efficiently but may yield optimistic estimates because temporally adjacent, overlapping windows can occur in different folds.
2. **Leave-One-Day-Out (LODO) CV** (temporal analysis): Data from each of the four collection days is held out in turn, with the model trained on the remaining three days. This evaluates out-of-distribution generalization across different recording sessions.

The stratified protocol measures performance when training and validation data are drawn from the same recording period, whereas LODO is the more stringent test of temporal transfer. Unless explicitly identified as LODO, the results below use stratified cross-validation. Final blind-test performance is computed by the challenge organizers and is not reported here.

4 Experiments and Results

4.1 Experimental Setup

The dataset comprises BLE RSSI recordings from 25 beacons deployed on the 5th floor of a nursing facility, collected over four days (April 10–13, 2023). A total of 23,905 timestamps are labeled with 22 room classes. A separate test set (April 14, 2023, approximately 5,700 timestamps) is withheld by the organizers for blind evaluation. Table 1 illustrates the class imbalance across rooms.

Table 1: Top-5 and bottom-5 rooms by sample count. Nurse station outnumbers the rarest room by $407\times$.

Room	Samples	Factor vs. rarest
Nurse station	9,361	$407\times$
Cafeteria	5,224	$227\times$
Kitchen	3,102	$135\times$
Hallway	2,890	$126\times$
Cleaning	2,105	$92\times$
Room 518	42	$1.8\times$
Room 517	42	$1.8\times$
Room 510	37	$1.6\times$
Room 505	44	$1.9\times$
Room 516	23	$1\times$

The Random Forest is configured with 100 trees, a maximum depth of 20, and `balanced_subsample` class weighting. These settings were fixed without a systematic hyperparameter search.

4.2 Overall Performance

Table 2 compares the five evaluated configurations under stratified 5-fold cross-validation.

Table 2: Performance comparison across five configurations. V3 (SG + class weight) is selected as the final model.

Configuration	W-F1	M-F1
V1: 17-class, no weight, no SG	0.8315	0.8193
V2: 22-class, class weight	0.8246	0.8191
V3: 22-class, SG, class weight	0.8351	0.8336
V3+4s: V3 + adaptive buffer 4s	0.8347	0.8350
V4: + relabeling (cosine)	0.8171	0.7713

V3 (SG + class weighting) attains the highest weighted F1-score, 0.8351, and a macro F1-score of 0.8336. Relative to V2, adding SG filtering increases weighted F1 from 0.8246 to 0.8351

and macro F1 from 0.8191 to 0.8336. V3+4s attains the highest macro F1-score, 0.8350, but its weighted F1-score is slightly lower (0.8347). We retain V3 as the primary configuration because it gives the best weighted score without the additional adaptive buffer. Relabeling (V4) reduces both aggregate metrics.

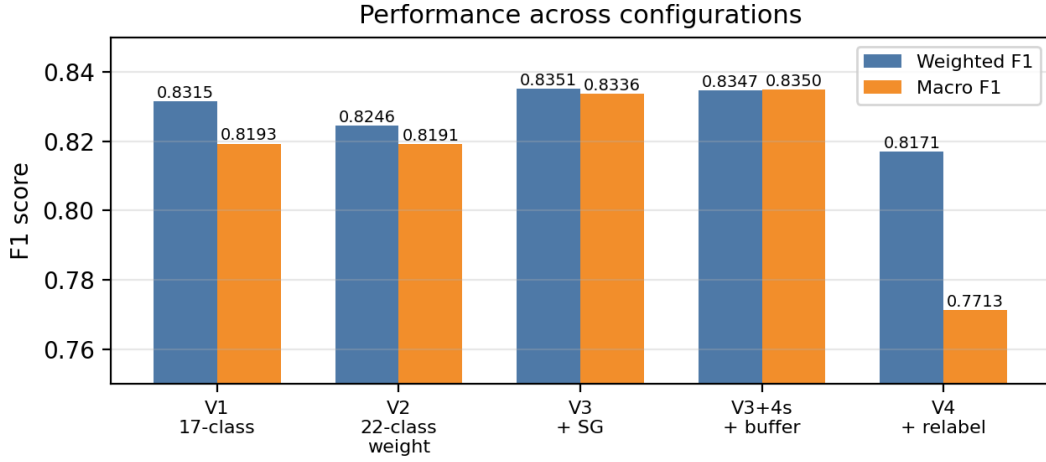


Figure 1: Weighted and macro F1-scores across configurations. V3 and V3+4s obtain the highest aggregate scores, whereas V4 performs worse on both metrics.

Figure 2 and Table 3 report the per-class results. All 22 classes exceed 0.65 F1 under stratified cross-validation, and most exceed 0.80. These results should be interpreted together with the lower LODO scores in Section 4, which show that within-period class performance does not directly translate to cross-day robustness.

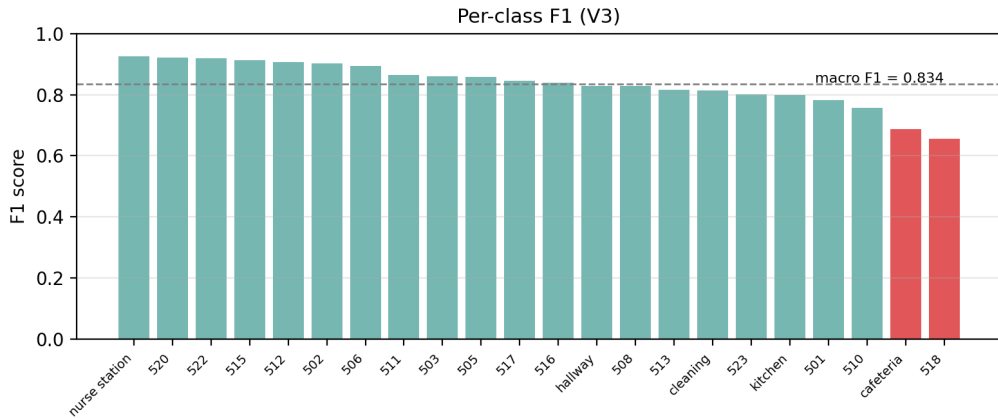


Figure 2: Per-class F1-scores for V3 under stratified 5-fold cross-validation. Classes below 0.75 are highlighted.

Table 3: Per-class F1-scores for V3 (SG + class weighting).

Class	F1	Class	F1
Nurse station	0.926	522	0.919
Kitchen	0.799	511	0.864
Cafeteria	0.688	502	0.903
Hallway	0.829	508	0.829
Cleaning	0.815	501	0.783
523	0.802	503	0.860
513	0.817	517	0.844
520	0.922	505	0.857
512	0.908	518	0.656
506	0.894	510	0.757
515	0.914	516	0.839

4.3 Overfitting Analysis

Figure 3 compares the train-validation gap across three imbalance strategies. SMOTE produces the largest gap (0.1459), whereas class weighting yields the smallest gap (0.0997). This result suggests that class weighting is less prone to overfitting than the evaluated synthetic-oversampling configuration.

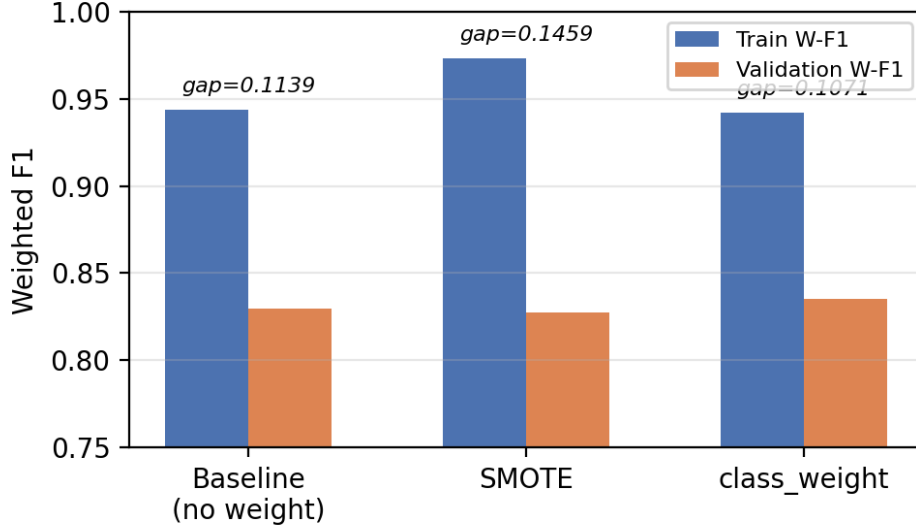


Figure 3: Train-validation gap across imbalance strategies. SMOTE has the largest gap, whereas class weighting has the smallest.

Figure 4 shows the confusion matrix of V3. Misclassifications are concentrated between adjacent rooms (e.g., room 513/515), reflecting the spatial ambiguity of RSSI fingerprints, while cross-zone errors remain negligible.

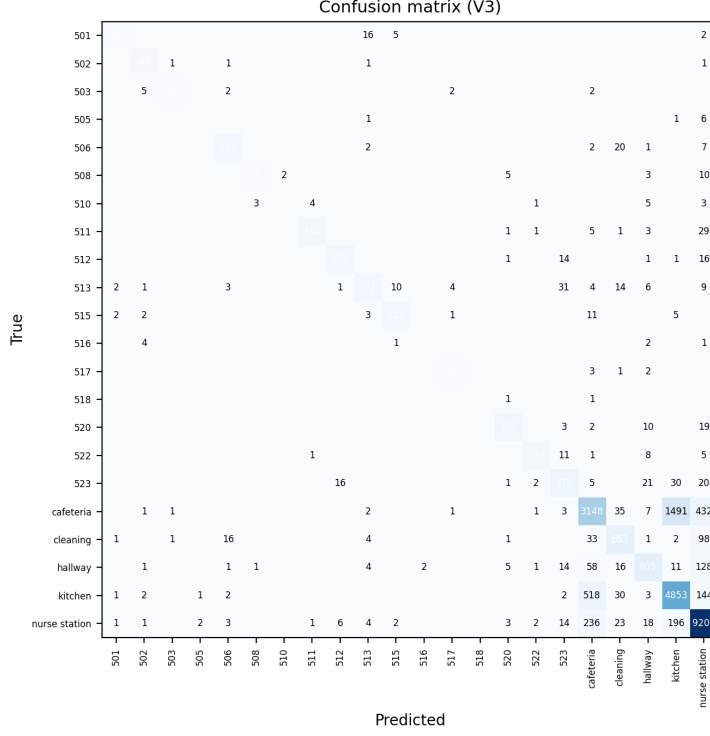


Figure 4: Confusion matrix of the V3 model (5-fold CV). Off-diagonal mass concentrates in spatially adjacent rooms.

4.4 Temporal Robustness

To assess generalization across recording sessions, we apply Leave-One-Day-Out CV:

Table 4: Leave-One-Day-Out cross-validation results. Lower scores reflect temporal RSSI variation across days.

Held-out day	Apr 10	Apr 11	Apr 12	Apr 13
W-F1	0.585	0.673	0.513	0.628

The mean LODO weighted F1-score of 0.600 is substantially lower than the stratified-CV result of 0.835. The gap indicates temporal distribution shift across recording days. Two plausible contributors are:

1. **Hardware latency:** The challenge documentation reports beacon–receiver delays of up to 60 seconds [10]. Such delays can cause short visits to be incompletely represented and may introduce label–signal misalignment.
2. **Environmental dynamics:** Day-level shifts in device orientation, body shadowing, and multipath fading alter RSSI distributions across recording sessions [2].

The current experiment does not isolate the causal contribution of latency and environmental variation. Nevertheless, the LODO results show that temporal generalization remains the principal limitation of the evaluated pipeline and motivates longer-term data collection and domain-adaptation methods.

5 Conclusion

This paper presented a Savitzky–Golay-filtered, class-weighted Random Forest pipeline for BLE-based indoor localization on a 22-room nursing floor. Under stratified 5-fold cross-validation, V3 achieves a weighted F1-score of 0.8351 and a macro F1-score of 0.8336 without synthetic data generation.

Our validation combines stratified and Leave-One-Day-Out protocols. The contrast between the stratified result and the mean LODO weighted F1-score of 0.600 shows that the former should be interpreted as an in-distribution estimate rather than evidence of cross-day robustness. Overlapping temporal windows may further make randomly stratified folds optimistic.

Future work includes collecting longer-term datasets for improved temporal robustness, exploring lightweight deep learning architectures, and developing domain adaptation methods to mitigate day-level RSSI shifts without requiring additional sensors.

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References

- [1] R. Faragher and R. Harle, “Location fingerprinting with Bluetooth low energy beacons,” *IEEE Journal on Selected Areas in Communications*, vol. 33, no. 11, pp. 2418–2428, 2015.
- [2] S. J. Halder, P. Giri, and W. Kim, “Advanced smoothing approach of RSSI and LQI for indoor localization system,” *International Journal of Distributed Sensor Networks*, vol. 11, no. 5, p. 195297, 2015.
- [3] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, “SMOTE: Synthetic minority over-sampling technique,” *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [4] A. Ayub, Z. Z. Abidin, A. Alhammadi, N. F. Soliman, M. A. Khan, and N. B. Ghazali, “Comparative analysis of machine learning algorithms for BLE-based indoor localization system,” *IEEE Access*, vol. 13, pp. 167 120–167 138, 2025.
- [5] P. J. García-Paterna, A. S. Martínez-Sala, and J. C. Sánchez-Aarnoutse, “Empirical study of a room-level localization system based on Bluetooth low energy beacons,” *Sensors*, vol. 21, no. 11, p. 3665, 2021.
- [6] C. A. Garcia and S. Inoue, “Relabeling for indoor localization using stationary beacons in nursing care facilities,” *Sensors*, vol. 24, no. 2, p. 319, 2024.
- [7] X. Hou, T. Arslan, and J. Gu, “Indoor localization for Bluetooth low energy using wavelet and smoothing filter,” in *Proc. Int. Conf. Localization GNSS (ICL-GNSS)*, 2017, pp. 1–6.
- [8] A. Savitzky and M. J. E. Golay, “Smoothing and differentiation of data by simplified least squares procedures,” *Analytical Chemistry*, vol. 36, no. 8, pp. 1627–1639, 1964.
- [9] L. Wang, S. Shang, and Z. Wu, “Research on indoor 3D positioning algorithm based on WiFi fingerprint,” *Sensors*, vol. 23, no. 1, p. 153, 2023.
- [10] ISAS 2026 Challenge, “Challenge tutorial – july 25, 2026,” 2026, canva presentation, mentioned beacon–receiver latency up to 60 seconds.